

ONLINE RETAIL ANALYSIS REPORT

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Programme: TaTa (Forgea)

Dataset: 541,910 Sales

Tools: Excel

EXECUTIVE SUMMARY

\$10.4M

Total Revenue

5.6M

Quantity Sold

\$521.20

AOV

375

Total Customer

38

Total Country

This report presents a data-driven analysis of the Online Retail Store's sales performance for the period 2010 to 2011. The dataset was cleaned to remove invalid entries including negative quantities, zero value in unit prices, missing customer records and unknown descriptions to ensure data integrity. Four key business questions were analysed using pivot tables and visualisations in Excel. The analysis revealed that revenue peaked in November 2011 at \$1.5M driven by pre-Christmas seasonal demand. The Netherlands emerged as the strongest international market by both revenue and quantity sold. Customer 14646 was identified as the highest revenue generating customer at \$280.2K. Regional demand analysis confirmed European countries as the dominant market outside the United Kingdom, with the Netherlands, EIRE and Germany leading in product demand. These findings provide actionable insights to support the CEO and CMO in making informed decisions around forecasting, customer retention and international market expansion.

PROBLEM STATEMENT

The CEO and CMO of the Online Retail Store require data driven insights to support strategic business decisions around revenue forecasting, international market expansion and customer retention. The existing transactional dataset for the period 2010 to 2011 contains data quality issues including negative quantities, zero value in unit prices, missing customer records and unknown descriptions that must be addressed before meaningful analysis can be conducted. This report aims to clean the dataset, analyse key business questions and present visual insights that enable both stakeholders to make informed decisions on seasonal revenue trends, top performing markets, highest value customers and regions with the greatest product demand.

DATA OVERVIEW

The dataset used for this analysis is the Online Retail Store transactional data spanning December 2010 to December 2011. The dataset contains 541,910 rows and 8 variables namely Invoice Number, Stock Code, Description, Quantity, Invoice Date, Unit Price, Customer ID and Country. The data captures customer transactions across multiple countries with the United Kingdom representing the majority of sales. Prior to analysis the dataset underwent a thorough cleaning process to ensure data integrity and accuracy.

Data Dictionary

Variable	Type	Description	Used in Analysis
Invoice No	INT	Unique identifier assigned to each transaction or order	Yes
Stock Code	INT	Unique identifier assigned to each product in the inventory	No
Description	Varchar	Name or description of the product purchased	Yes
Quantity	INT	Number of units purchased per transaction	Yes
Invoice Date	Datetime	Date and time the transaction was recorded	Yes
Unit Price	Decimal	Price per unit of the product in GBP	Yes
Customer ID	INT	Unique identifier assigned to each customer	Yes
Country	Varchar	Country where the customer is located	Yes

Data Quality Issues & Decisions

Issue	Volume	Decision
Duplicate Value	1% (5268)	Remove duplicate by all column
Anomalies	2% (10589)	Exclude them out of the data
Missing value	(5)	Drop the missing value rows because it is not relevant to the analysis
Datatype	All column	Give each column the appropriate data type
Unspecified Name	-	Exclude the row which Unspecified was mention because no price and customer ID was given

METHODOLOGY

A five-stage pipeline using Excel was adopted to ensures reproducibility and separation of raw from analytical data.

Stage	Tool	Output	Rationale
Ingestion	Excel	541,910 rows with 8 variables	Confirmed dataset is complete and ready for exploration
Exploration	Excel	Identified datatypes, variable structures and potential quality issues across all column	Recognised presence of missing value, anomalies and unknown description
Cleaning & Transformation	Excel	Data check column created to exclude invalid rows covering negative quantities, zero unit prices, Revenue column created using Quantity x Unit Price	Ensured only valid and complete transactions are used for analysis to prevent skewed results
Analysis	Excel – Pivot table	Summarised data by Month, Country and Customer ID to answer all four stakeholder questions	Filtered 2011 only for Q1, excluded United Kingdom for Q2, Q3 and Q4
Visualization	Excel	Line Chart for Q1, Clustered Column Chart for Q2 and Q3, Map Chart for Q4	Display insights through charts

KEY FINDINGS

1. Overall Performance Overview

The Online Retail Store recorded a total revenue of \$9.6M for the year 2011 across multiple international markets. The dataset captured transactions from over 36 countries with

European markets dominating both revenue and quantity sold outside the United Kingdom. The analysis identified strong seasonal patterns, high value customer segments and regions with significant expansion potential, providing a comprehensive view of the business performance across the period.

2. Monthly Revenue Trend (2011)

The revenue for 2011 shows a generally upward trajectory throughout the year, starting at \$682.2k in January and peaking at \$1.5M in November. The business experiences strong seasonal demand in Q4, particularly in October and November, which is likely driven by pre-Christmas consumer purchasing behaviour. Revenue remained relatively stable between \$682.2K to \$702.0K from January to July before accelerating significantly in the latter half of the year.

Note: December revenue of \$417.9k appears significantly lower than the preceding months and should be interpreted with caution as the data for this month may be incomplete and therefore does not reflect the true performance for that period.

Implication: The company should prioritise stock availability and increase marketing efforts from August onwards to capitalise on the Q4 seasonal surge and maximise revenue during peak months.

3. Top 10 Revenue Generating Countries and Quantity Sold

The Netherlands emerged as the highest revenue generating country outside the United Kingdom at approximately \$285.4K in revenue and 200.4K units sold, followed by EIRE and Germany as the second and third strongest markets. A positive correlation is observed between quantity sold and revenue across all top countries, suggesting that higher purchase volume directly drive revenue growth. Japan recorded the lowest performance among the top 10 countries with \$37.4K in revenue and 26.0K units sold.

Implication: The CMO should prioritise strengthening marketing and distribution efforts in the Netherlands and EIRE as these European markets demonstrate the strongest commercial performance outside the United Kingdom.

4. Top 10 Customers by Revenue

Customer 14646 generated the highest revenue among the top 10 customers at approximately \$280.2K, followed closely by Customer 18102 at \$259.7K and Customer 17450 at \$194.4K. The chart shows gradual decline in revenue from the highest to the lowest performing customer, with Customer 12346 recording the lowest revenue among the top 10 at \$77.2K.

Implication:

The CMO should prioritise retaining and engaging the top revenue generating customers such as Customer 14646 and 18102 through targeted loyalty programs and personalized offerings to sustain and grow their contribution to overall revenue.

5. Countries with high demand

The map chart reveals that European countries collectively demonstrate the highest demand for products, with the Netherlands recording the greatest quantity sold at 200.4K units, followed by EIRE at 147.0K and Germany at 119.1K. France and Australia also show considerable demand at 112.1K and 83.9K units respectively. Countries such as Canada, Poland and Isreal recorded the lowest demand figures, suggesting limited market penetration in those regions.

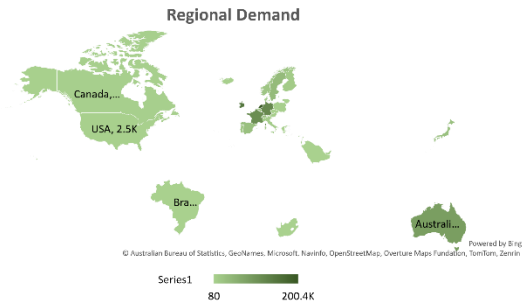
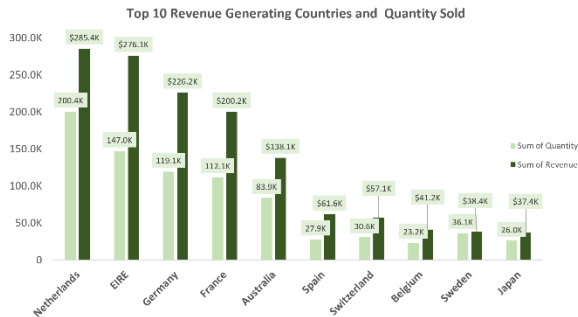
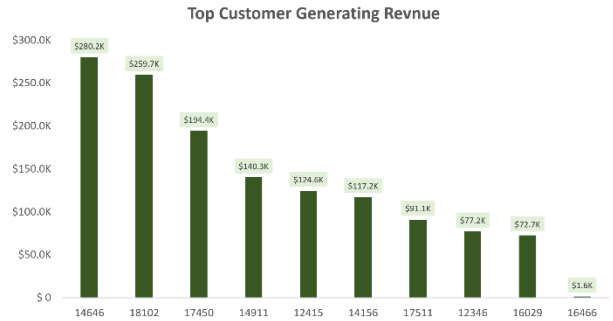
Implication:

The CEO should prioritise expansion efforts in high demand European markets such as the Netherlands, EIRE and Germany while developing targeted entry strategies for lower demand regions such as Asia and the Americas where significant growth opportunities exist.

Visual Insight & Chart

An interactive Excel dashboard was published comprising **2 pages** with dynamic slicers.

Page	Question Addressed	Visuals
Page 1 — Question 1	CEO want to know: What is the monthly revenue trend for 2011 and what seasonal patterns can be observed?	A line chart to show monthly trend for 2011
Page 2 — Question 2	CMO want to know: Which are the top 10 revenue generating countries excluding the United Kingdom and what quantity was sold?	A stacked column showing top 10 countries generating revenue and quantity sold
Page 3 — Question 3	CMO want to know: Who are the top 10 customers by revenue ranked from highest to lowest?	A column chart showing the top 10 customers generating revenue
Page 4 — Question 4	CEO want to know: Which regions globally have the highest product demand excluding the United Kingdom?	A map showing the country with highest demand



RECOMMENDATION

1. Monthly Revenue Trend

The Online Retail Store should increase inventory levels and marketing spend ahead of Q4, particularly from August onwards, to maximise revenue during the peak seasonal period and ensure sufficient stock availability to meet heightened consumer demand.

2. Top 10 Revenue Generating Countries

The Online Retail Store should strengthen its commercial presence in the Netherlands and EIRE by increasing targeted marketing campaigns and optimising distribution channels to sustain and grow revenue in these high performing European markets.

3. Top 10 Customers by Revenue

The Online Retail Store should introduce a dedicated loyalty and engagement programme targeting its highest value customers, particularly Customer 14646 and 18102, to reinforce retention and encourage continued high value purchasing behaviour.

4. Regional Demand

The Online Retail Store should develop a phased market expansion strategy focused on high demand European markets while simultaneously exploring growth opportunities in

underpenetrated regions such as Asia and the Americas to diversify its international revenue base.

CONCLUSION

The analysis of the Online Retail Store's transactional data reveals strong seasonal revenue performance in Q4 2011, with European markets driving the majority of international sales. The Netherlands, EIRE and Germany consistently emerged as the highest performing markets outside the United Kingdom across both revenue and quantity sold. A concentrated base of high value customers presents a significant retention opportunity for the CMO, while underpenetrated regions such as Asia and the Americas offer meaningful expansion potential for the CEO. Overall the findings suggest that with targeted marketing, strategic expansion and proactive customer engagement, the Online Retail Store is well positioned to sustain and grow its revenue in the coming year.